

ZENOFEST 2K26

PROJECT EXPO SUBMISSION

Team ID: ZFPE1	Team Name: EcoSense
Project Title: EcoSense - Smart Waste Management for College Campuses	Leader Name: Vaishnavi
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SUBMISSION DATA

Development Domain: IoT / Smart Campus

Problem Statement: College campuses produce hundreds of kilograms of waste daily, but bin collection schedules are fixed regardless of actual fill levels.

Proposed Solution: IoT-enabled smart bins with ultrasonic fill sensors that report fill levels in real time.

Technologies Used: Arduino Uno, HC-SR04, ESP8266, Python, React

Innovation / Uniqueness: Low-cost off-the-shelf parts with predictive routing.

Implementation: Phase 1: prototype. Phase 2: field pilot. Phase 3: campus scale.

Target Users: Campus maintenance teams and facility managers.

Future Scope: Reward system, image classification, city integration.

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